

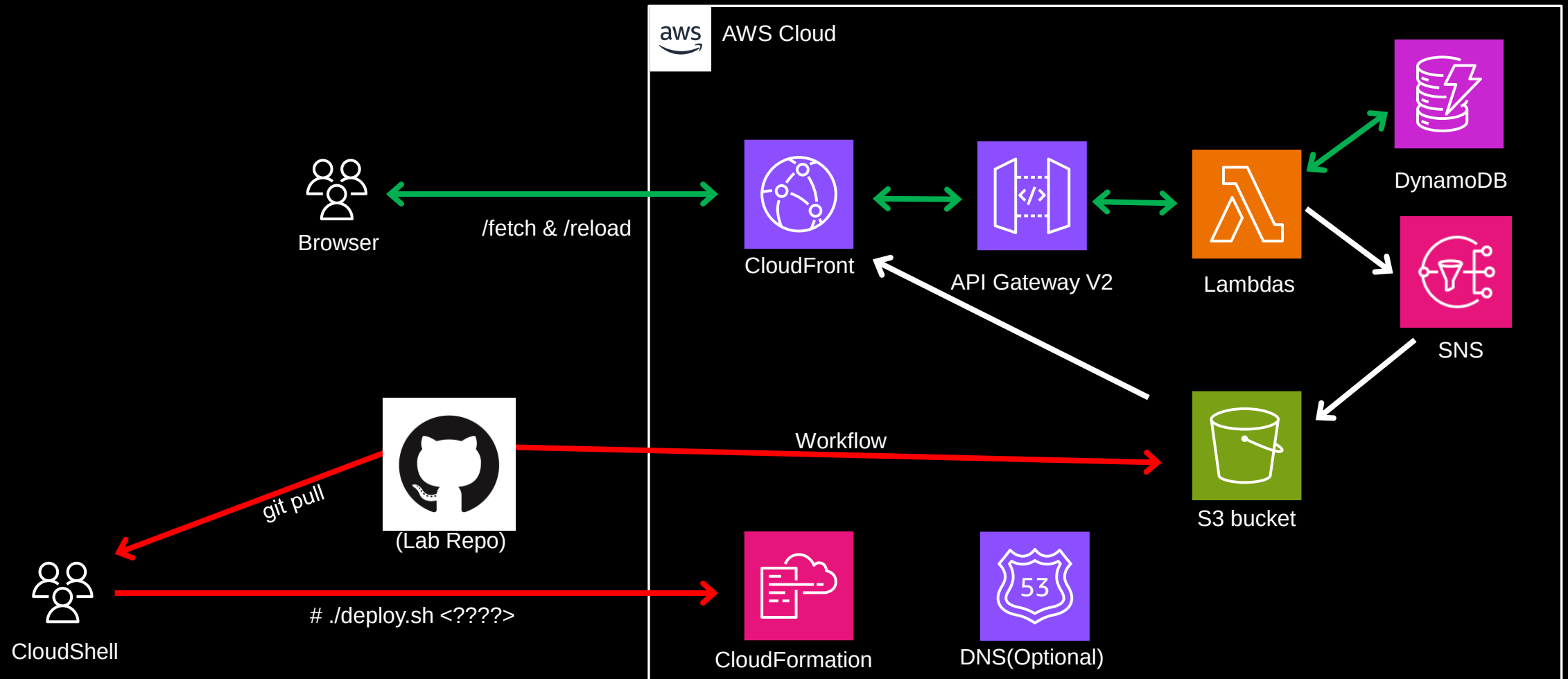
Eeny-Meany-Miny-Moe

Ken Graf

**AWS solution to pick next
student to present**

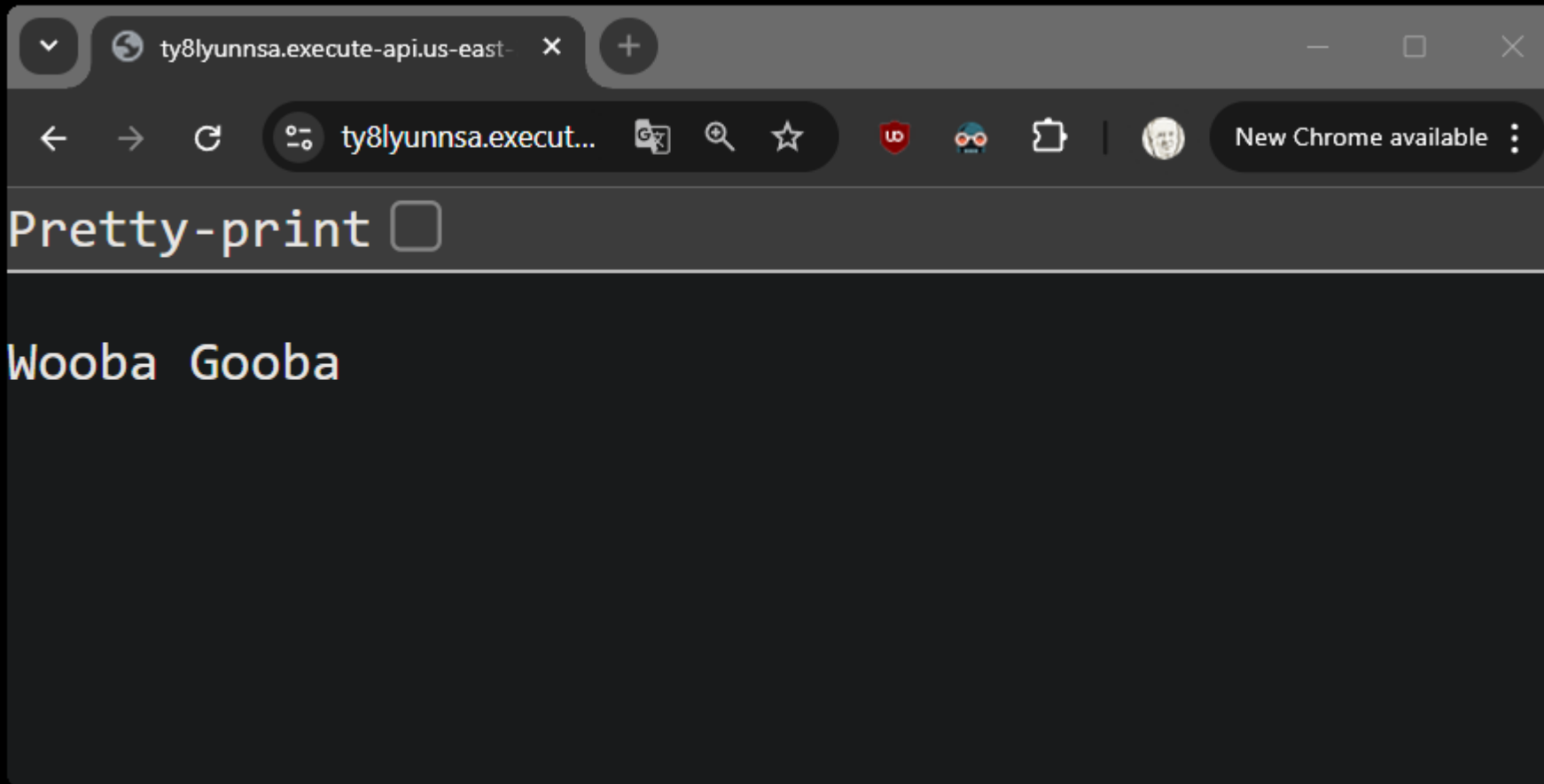


Eeny-Meany-Miny-Moe Components



PROJECT DEMO: No Fancy GUI

- Make a “curl” request get the next name



SECURITY #10: Develop incident management plans

- Allow for deployment to another region within 1 minute after regional AWS failure
- RECOMMENDATIONS:
 - Drop US-EAST-2 dependencies
 - Replicate S3 into us-west-2
 - Create playbook to meet the minute 1-minute goal

RELIABILITY #13: Use defined recovery strategies to meet the recovery objectives

- Programmatically repopulate the database on failures
 - Items exhaustion due to over selection
 - Deletion of all items
 - Table deletion
- **RECOMMENDATION: Add recovery code to Lambda function**

OPERATIONAL EXCELLENCE #8: Create actionable alerts

- **RECOMMENDATION:** Python code for DynamoDB health check.

```
def main():
    topic_name = "AWSHealthNotifications"
    rule_name = "AWSHealthDynamoDBAlerts"
    email = "kmh722@unh.edu"

    print("Creating SNS topic...")
    topic_arn = create_sns_topic(topic_name)
    print(f"SNS Topic ARN: {topic_arn}")

    print("Subscribing email to SNS...")
    subscribe_to_sns(topic_arn, email)

    print("Setting SNS policy...")
    set_sns_policy(topic_arn)

    print("Creating EventBridge rule...")
    create_eventbridge_rule(rule_name, topic_arn)
    print("Setup complete. Please confirm SNS email subscription.")
```

COST #6: Perform cost modeling

- Move to Python from Node.js?

	128MB	256MB	512MB	1024MB
Node.js v20	470ms	421ms	322ms	289ms
Python 3.13	68m	28ms	18ms	17ms



- Used Apache Bench after lambda warmup:

ab -c 1000 -n 100000 -l {url}

- Python @ 258MB: 96% cost savings vs. node.js @ 512MB

- **RECOMMENDATION: Move to Python**

PERFORMANCE #1: Factor cost into architectural decisions

- Drop ApiGatewayV2?
- Not using Authorization, caching, Service integration.
- Tested user response times with Apache Bench.
ab -c 1000 -n 100000 -l {url}
- ApiGateway: ~40ms delay added, better concurrency handling
- Roundtrips from Australia to us-east-2: ~900ms
- **RECOMMENDATION: Simplify the deployment, drop gateway**

Eeny-Meany-Miny-Moe REVISED

